Computation Graph

Operation of a neural network consist of 2 stages:

① Forward Propagation to compute the output.

② Backward Propagation to update the weights to minimise error.

Computation graph explain why we have 2 steps.

Ex: Let us compute the function:

$$J(a,b,c) = 3(a+b+c)$$

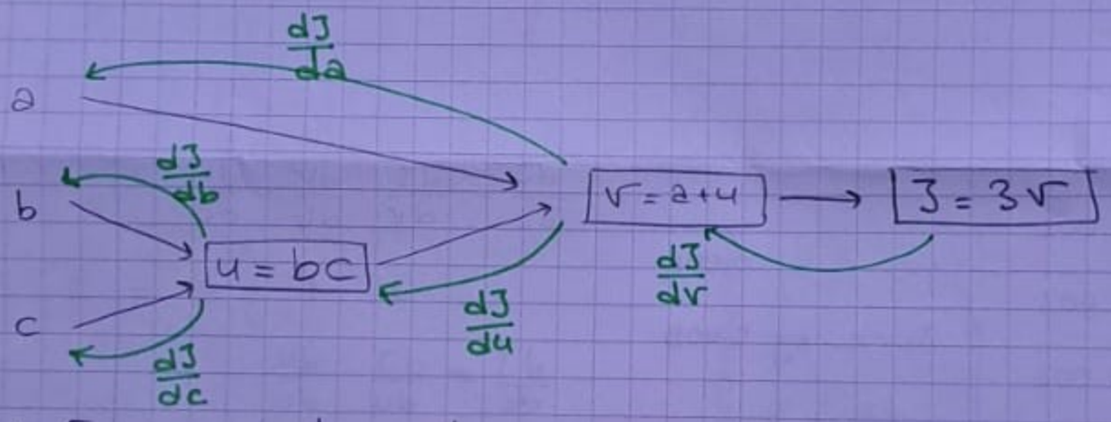
Computation consists of 3 steps

$$u = bc$$

$$v = a + u$$

$$J = 3v$$

Draw 3 steps in computation graph.



if $a = 5$, $b = 3$, $c = 2$ $\Rightarrow$ $u = bc = 6$, $v = a+u = 5+6 = 11$, $J = 3v = 3 \cdot 11 = 33$

① $\frac{dJ}{dv} = ?$

$$\frac{dJ}{dv} = \frac{d(3v)}{dv} = 3$$

$J = 3v$

$v = 11 \xrightarrow[\text{a little}]{\text{incr.}} 11.001$

$J = 33 \longrightarrow 33.003$

J changes by 3×0.001 $\quad \frac{dJ}{dv} = 3$

② $\frac{dJ}{da} = ?$

$a = 5 \xrightarrow[\text{a little}]{\text{incr.}} 5.001$

$v = 11 \longrightarrow 11.001$

$J = 33 \longrightarrow 33.003$

if we increase a by 0.001, J incr. by 0.003.

$\frac{dJ}{da} = \frac{0.003}{0.001} = 3$

$$\frac{dJ}{da} = \underbrace{\frac{dJ}{dv}}_{3} \cdot \underbrace{\frac{dv}{da}}_{1} \quad \Leftarrow \text{chain rule.}$$

$\frac{dJ}{da} = 3$; step ② of backprop.

③ $\frac{dJ}{du} = ?$

$u = 6 \longrightarrow 6.001$

$v = 11 \longrightarrow 11.001$

$J = 33 \longrightarrow 33.003$

incr. u by 0.001
J increase 0.003

$\frac{dJ}{du} = 3$

$\frac{dJ}{du} = \frac{dJ}{dv} \cdot \frac{dv}{du}$ ($\frac{dJ}{dv} = 3$, $\frac{dv}{du} = 1$)

$\frac{dJ}{du} = 3$

④ $\frac{dJ}{db} = ?$

$b = 3 \longrightarrow 3.001$

$u = 6 \longrightarrow 6.002$

$v = 11 \longrightarrow 11.002$

$J = 33 \longrightarrow 33.006$

increase b by 0.001
J incr. by 0.006

$\frac{dJ}{db} = 6$

$\frac{dJ}{db} = \frac{dJ}{dv} \frac{dv}{du} \frac{du}{db}$

$= 3 \cdot 1 \cdot 2 = 6$

$\frac{dJ}{db} = \frac{dJ}{du} \frac{du}{db}$

$= 3 \cdot 2 = 6$

⑤ $\frac{dJ}{dc} = ?$

$c = 2 \longrightarrow 2.001$

$u = 6 \longrightarrow 6.003$

$v = 11 \longrightarrow 11.003$

$J = 33 \longrightarrow 33.009$

inc. c by 0.001
J inc. by 0.009

$\frac{dJ}{dc} = 9$

$\frac{dJ}{dc} = \frac{dJ}{dv} \cdot \frac{dv}{du} \cdot \frac{du}{dc}$

$= 3 \cdot 1 \cdot 3 = 9$

$\frac{dJ}{dc} = \frac{dJ}{du} \cdot \frac{du}{dc}$

$= 3 \cdot 3 = 9$